

Chapter 12: Stoichiometry

Question	Answer	Justification / Calculation
1	B	The coefficients represent particles: 2 molecules of hydrogen react with 1 molecule of oxygen to form 2 molecules of water.
2	C	The coefficients represent moles: 1 mole of nitrogen reacts with 3 moles of hydrogen to form 2 moles of ammonia.
3	B	The mass of the reactants (4.0 g H ₂ + 32.0 g O ₂) equals the mass of the products (36.0 g H ₂ O).
4	B	A mole ratio is a conversion factor derived directly from the coefficients of a balanced chemical equation.
5	B	Using the 3:2 molar ratio from the balanced equation, $6.0 \text{ mol H}_2 \times (2 \text{ mol NH}_3 / 3 \text{ mol H}_2) = 4.0 \text{ mol NH}_3$.
6	C	Using the 3:2 ratio from the equation, 4.00 mol CO ₂ requires 6.00 mol O ₂ . The molar mass of O ₂ is 32.0 g/mol, so $6.00 \text{ mol} \times 32.0 \text{ g/mol} = 192 \text{ g}$.
7	D	$48.0 \text{ g CH}_4 \div 16.0 \text{ g/mol} = 3.0 \text{ mol CH}_4$. The ratio of CH ₄ to O ₂ is 1:2, requiring 6.0 mol O ₂ . $6.0 \text{ mol} \times 32.0 \text{ g/mol} = 192 \text{ g}$.
8	B	Coefficients specifically indicate the relative number of moles or particles for reactants and products.
9	B	$46.0 \text{ g Na} \div 23.0 \text{ g/mol} = 2.0 \text{ mol Na}$. The Na to H ₂ O ratio is 2:2, meaning 2.0 mol H ₂ O are needed. $2.0 \text{ mol} \times 18.0 \text{ g/mol} = 36.0 \text{ g}$.
10	B	Stoichiometry is the portion of chemistry that calculates numerical quantities in chemical equations.
11	A	The limiting reagent dictates the maximum possible amount of product that can be formed.
12	C	Identification requires expressing quantities as moles and comparing them against the balanced mole ratios.
13	B	9.0 mol H ₂ requires 3.0 mol N ₂ , leaving N ₂ in excess, so H ₂ is limiting. $9.0 \text{ mol H}_2 \times (2/3) = 6.0 \text{ mol NH}_3$.
14	C	Percent yield = (Actual Yield / Theoretical Yield) $\times$ 100. $(22.4 \text{ g} / 28.0 \text{ g}) \times 100 = 80.0\%$.
15	A	Actual yield is defined as the amount formed during laboratory execution.
16	C	The limiting reagent is the reactant that is entirely consumed first.
17	C	Percent yield is the ratio of actual yield to theoretical yield, acting as a measure of

Question	Answer	Justification / Calculation
		reaction efficiency.
18	D	Theoretical yield represents the maximum calculated product from a given reactant.
19	B	A mole ratio is a conversion factor derived from equation coefficients.
20	C	Actual yield is the amount of product formed in the laboratory.
21	G	Theoretical yield represents the calculated maximum product.
22	E	The limiting reagent is used up first and determines the product amount.
23	A	Stoichiometry deals with numerical relationships and calculation of quantities.
24	F	Percent yield measures efficiency as a ratio of actual to theoretical yield.
25	D	Excess reagent is the reactant remaining after the limiting reagent is consumed.

Chapter 14: The Behavior of Gases

Question	Answer	Justification / Calculation
1	B	Compression specifically reduces the space between the gas particles.
2	C	Increased kinetic energy at a constant volume causes more frequent and forceful particle collisions.
3	A	Moving to freezing temperatures lowers kinetic energy, exerting less pressure on the balloon walls, causing volume to decrease.
4	C	Adding 4.0 moles to the existing 2.0 moles brings the total to 6.0 moles (tripling the amount). At constant volume and temperature, pressure triples.
5	C	Volume is reduced by a factor of 5 (10.0 L to 2.0 L). Following Boyle's Law, pressure quintuples.
6	B	Removing gas reduces the total number of particle collisions against the container walls.
7	B	Charles's Law mathematically relates volume and temperature: $V_1/T_1 = V_2/T_2$.
8	B	Applying Gay-Lussac's law, halving the temperature (400 K to 200 K) halves the pressure to 2.00×10^2 kPa.
9	A	Using Boyle's Law ($P_1V_1 = P_2V_2$), $V_1 = (1.50 \times 10^2 \times 4.00) / (2.00 \times 10^2) = 3.00$ L.

Question	Answer	Justification / Calculation
10	C	Using Boyle's Law, as pressure drops by a factor of 4 (100.0 kPa to 25.0 kPa), volume increases by a factor of 4 ($2.0 \text{ L} \times 4 = 8.0 \text{ L}$).
11	C	Using Charles's Law, tripling the volume (400 mL to 1200 mL) requires tripling the temperature ($300 \text{ K} \times 3 = 900 \text{ K}$).
12	C	The combined gas law formula is $(P_1V_1)/T_1=(P_2V_2)/T_2$.
13	B	Using Gay-Lussac's Law, $P_2 = P_1 \times (T_2/T_1) = 1.50 \times 10^2 \times (400/300) = 2.00 \times 10^2 \text{ kPa}$.
14	B	Gay-Lussac's law assumes constant volume, so the V_1 and V_2 terms cancel out of the combined gas law.
15	C	An ideal gas is defined by having no volume, perfectly elastic collisions, and no molecular attractions.
16	A	Truly ideal gases do not exist because real particles have finite volume and experience attractive forces.
17	C	Using $PV=nRT$, $V = (2.00 \times 8.31 \times 300) / 100.0 = 49.86 \text{ L}$ (rounds to 49.9 L).
18	B	$14.0 \text{ g N}_2 = 0.500 \text{ mol}$. $V = (0.500 \times 8.31 \times 300) / 50.0 = 24.93 \text{ L}$.
19	B	$8.00 \text{ g He} = 2.00 \text{ mol}$. $P = (2.00 \times 8.31 \times 400) / 4.00 = 1662 \text{ kPa}$ or $1.66 \times 10^3 \text{ kPa}$.
20	C	$n = (200.0 \times 1.00 \times 10^2) / (8.31 \times 300) = 8.02 \text{ mol}$. $8.02 \text{ mol} \times 39.95 \text{ g/mol} = 320 \text{ g}$ or $3.2 \times 10^2 \text{ g}$.
21	B	Dalton's law states total pressure is the sum of individual partial pressures at constant volume and temperature.
22	A	Helium (4.00 g/mol) effuses faster than Neon (20.18 g/mol). Ratio = $\sqrt{(20.18/4.00)} \approx 2.25$.
23	B	Molecular nitrogen (28.0 g/mol) effuses faster than carbon dioxide (44.0 g/mol) due to lower molar mass.
24	C	Ratio = $\sqrt{(32/2)} = \sqrt{16} = 4$, yielding a 4:1 ratio.
25	A	Ratio = $\sqrt{(40/4)} = \sqrt{10} \approx 3.16:1$.
26	B	Compressibility is the measure of how volume decreases under pressure.
27	C	Boyle's law describes the inverse relationship between volume and pressure.
28	B	Charles's law states volume is directly proportional to Kelvin temperature at constant pressure.

Question	Answer	Justification / Calculation
29	B	$PV=nRT$ is known as the ideal gas law.
30	B	Diffusion is the tendency of molecules to move toward lower concentration areas.
31	D	Effusion is gas escaping through a tiny hole in its container.
32	A	Graham's law relates the rate of effusion to the inverse square root of molar mass.
33	C	Partial pressure is the individual contribution of a gas to total pressure.
34	A	Compressibility is how much volume decreases under pressure.
35	B	Boyle's law governs the inverse relationship of volume and pressure.
36	C	Charles's law governs the direct relationship of volume and temperature.
37	D	The combined gas law relates pressure, temperature, and volume.
38	E	Gay-Lussac's law governs the direct relationship of pressure and temperature.
39	F	The gas constant is represented by symbol R .
40	G	The gas law refers to the ideal gas relationship $PV=nRT$.
41/42	I	Diffusion represents molecules moving toward areas of lower concentration.
43	J	Effusion is escaping gas through a tiny hole.
44	K	Graham's law relates effusion rate to molar mass.
45	L	Partial pressure is a gas's contribution to total pressure.
N/A	H	Dalton's law of partial pressures equates total pressure to the sum of partial pressures.

Chapter 15: Water and Aqueous Systems

Question	Answer	Justification / Calculation
1	B	Strong hydrogen bonding creates the inward pull minimizing surface area.
2	C	Surface molecules are attracted to liquid below but not the air above.
3	B	Water's surface tension causes spherical drops and allows denser objects to float.
4	D	Surfactants interfere with hydrogen bonding, ultimately reducing surface tension.

Question	Answer	Justification / Calculation
5	C	Hydrogen bonds physically hold water molecules to each other, restricting escape.
6	D	The slow evaporation rate is due to water's intrinsically low vapor pressure.
7	A	Ice forms a regular, open framework of hydrogen-bonded molecules.
8	D	If denser, ice would sink, causing aquatic bodies to freeze bottom-up and kill life.
9	C	Hydrogen bonds hold ice molecules in a regular, open structure.
10	D	An aqueous solution strictly defines water as the solvent.
11	B	Water acts as the dissolving solvent; sugar is the dissolved solute.
12	B	Nonpolar compounds lack charges and are unaffected by the electrostatic attraction of polar water.
13	A	The dissolved molecules and ions are smaller than standard filter pores.
14	C	Solvation is the process of cations and anions becoming surrounded by solvent molecules.
15	D	CH ₄ is nonpolar and will not appreciably dissolve in polar water.
16	B	Separation occurs because water is polar and gasoline is nonpolar.
17	D	Ions must be present in a solution to conduct an electric current.
18	B	Molten states free the ions to move toward an electrode.
19	D	A strong electrolyte is almost totally ionized in solution.
20	C	Water of hydration is water contained within the crystal structure of a substance.
21	C	The prefix "octa-" dictates 8 water molecules: Ba(OH) ₂ · 8H ₂ O.
22	D	Fe ²⁺ is Iron(II), and "hepta-" means 7: Iron(II) sulfate heptahydrate.
23	A	Heating to monohydrate removes 6 water molecules: MgSO ₄ · 7H ₂ O → MgSO ₄ · H ₂ O + 6H ₂ O.
24	D	Hygroscopic desiccants function by absorbing water vapor from the air.
25	B	They must be sealed because they rapidly absorb water vapor from ambient air.
26	A	Loss occurs when the hydrate's vapor pressure exceeds that of the ambient air.
27	C	Suspensions contain large particles that settle, while solution particles stay suspended.

Question	Answer	Justification / Calculation
28	C	Correct increasing size order: Solutions (smallest), colloids, suspensions (largest).
29	B	The Tyndall effect is the scattering of visible light by colloids and suspensions.
30	D	True solution molecules are too small to provide reflective surfaces.
31	B	Caused by collisions of dispersing medium molecules with dispersed colloidal particles.
32	D	Brownian motion and repulsion between like-charged ions keep them suspended.
33	A	Adding oppositely charged ions destroys the stability of the dispersion.
34	B	An emulsifier is added to stabilize a colloidal dispersion.
37	B	A surfactant is a wetting agent that reduces surface tension.
38	B	Solvation is when ions become surrounded by solvent molecules.
39	A	A strong electrolyte is almost totally ionized.
40	C	The Tyndall effect identifies the scattering of visible light.
41	D	Efflorescent hydrates lose water due to high vapor pressure.
Mat 1	D	Aqueous solutions have water as the solvent.
Mat 2	B	The solvent is the dissolving medium.
Mat 3	G	The solute is the substance being dissolved.
Mat 4	E	Water of hydration is water in a crystal structure.
Mat 5	I	A desiccant absorbs water vapor from the air.
Mat 6	A	A suspension has large particles that settle.
Mat 7	H	A colloid has intermediate particle sizes.
Mat 8	J	Brownian motion is caused by collisions of medium molecules.
Mat 9	C	An emulsifier makes a colloidal dispersion stable.
Mat 10	F	A surfactant reduces surface tension.

Chapter 16: Solutions

Question	Answer	Justification / Calculation
1	C	The solvent is the substance in which the solute is dissolved.
2	B	Random solvent collisions provide the force to overcome gravity.
3	D	Cooling a saturated solution lowers solubility, causing solute particles to crystallize.
4	C	No; undissolved solute forces excess to come out of a supersaturated solution.
5	C	Utilizing solubility curves (approx 222 g/100g water at 20°C), $5 \times 222 \text{ g} = 1110 \text{ g}$ or $1.12 \times 10^3 \text{ g}$.
6	B	Gas solubility is directly proportional to pressure, thus solubility increases with pressure.
7	B	Using Henry's Law ($S_1/P_1 = S_2/P_2$): $0.040/1.00 = x/0.50$, $x = 0.020 \text{ g/L}$.
8	C	Dilute and concentrated are relative, whereas molarity gives the exact moles per liter.
9	C	Molarity = Moles / Liters. $2.0 \text{ mol} / 0.500 \text{ L} = 4.0\text{M}$.
10	C	$M_1V_1 = M_2V_2$. $2.00 \times V_1 = 0.500 \times 250.0$. $V_1 = 62.5 \text{ mL}$.
11	B	Moles = $M \times V = 0.25 \times 2.0 = 0.50 \text{ mol}$. Mass = $0.50 \text{ mol} \times 40 \text{ g/mol} = 20 \text{ g}$.
12	B	$2.0\% = (\text{Mass} / 1500) \times 100$. Mass = $0.02 \times 1500 = 30 \text{ g}$.
13	B	$\%V/v = (40 \text{ mL} / 200 \text{ mL}) \times 100 = 20\%$.
14	B	Colligative properties depend only on the number of solute particles.
15	A	2.0M NaCl dissociates into 4.0M particles, while 1.0M MgCl ₂ yields 3.0M. Higher particle concentration yields higher boiling point.
16	B	CaCl ₂ forms 3 particles per unit (effective molality 3m); KCl forms 2 (effective molality 2m).
17	B	Lowered vapor pressure requires more energy to reach atmospheric pressure, increasing boiling point.
18	B	The salt acts as a solute to lower the freezing point of the ice-water mixture.
19	A	1M is defined per liter of solution; 1m is defined per 1000 g of solvent.
42	B	Immiscible describes liquids that are insoluble in one another.
43	C	Solubility is the amount that dissolves in a given quantity of solvent.
44	D	Henry's Law equates gas solubility directly to pressure.

Question	Answer	Justification / Calculation
45	A	A saturated solution holds the maximum amount of solute.
46	B	Mole fraction is the ratio of solute moles to total moles.
Mat 1	B	A concentrated solution contains a large amount of solute.
Mat 2	G	A dilute solution contains a small amount of solute.
Mat 3	D	A supersaturated solution contains more solute than it can theoretically hold.
Mat 4	E	Boiling-point elevation is the temperature difference relative to pure solvent.
Mat 5	F	Molality measures moles of solute per kilogram of solvent.
Mat 6	C	K _f is the change in freezing point for a 1-molal solution.
Mat 7	A	A colligative property depends only on particle number.

Chapter 17: Thermochemistry

Question	Answer	Justification / Calculation
1	B	Energy is neither created nor destroyed in chemical/physical processes.
2	C	Heat flows directly from the warmer object (aluminum) to the cooler one (water).
3	B	Potential energy is stored due to a substance's chemical composition.
4	B	Heat capacity relies on mass and chemical composition.
5	B	1000 chemistry calories (cal) equals 1 dietary Calorie (Cal).
6	C	$350 \text{ cal} \times 4.184 \text{ J/cal} = 1464.4 \text{ J}$ or $1.46 \times 10^3 \text{ J}$.
7	B	Defining boundaries accurately measures heat flow between them.
8	C	Heat flow out of a system is negative; flow into the system is positive.
9	B	Freezing cold means it absorbed heat (endothermic); the beaker and air are surroundings.
10	B	Evaporating liquid nitrogen requires absorbing heat (endothermic).
11	C	Enthalpy represents heat change occurring at constant pressure.
12	B	Calorimeters are designed to measure enthalpy changes.

Question	Answer	Justification / Calculation
13	B	Incomplete mixing and heat loss to ambient air are sources of error.
14	C	A bomb calorimeter measures heat released at constant volume.
15	B	Standard conditions are 101.3 kPa (1 atm) and 25°C.
16	C	A thermochemical equation explicitly includes the heat released or absorbed.
17	A	Incoming heat fuels the phase change (melting ice) rather than increasing kinetic energy.
18	B	Freezing loses heat: $2.00 \text{ mol} \times (-6.01 \text{ kJ/mol}) = -12.0 \text{ kJ}$.
19	D	The reaction has a positive ΔH (25.7 kJ/mol), meaning it absorbs heat (endothermic) and decreases water temperature.
20	B	Hess's law allows enthalpy calculation from the known enthalpies of multiple steps.
21	D	Reverse reaction changes sign to -66.4 kJ for 2 mol; dividing by 2 for 1 mol yields +33.2 kJ, which is endothermic.
22	A	Summing the steps directly gives $-400.0 \text{ kJ} + (-250.0 \text{ kJ}) = -650.0 \text{ kJ}$.
23	C	The standard heat of formation for a free element in its standard state is exactly zero.
24	B	Greater heat release (more negative) indicates higher stability relative to its constituent elements.
25	B	Specific heat describes the heat needed to raise 1 gram by 1°C.
26	C	Thermochemistry studies energy changes during reactions.
27	A	Enthalpy is the heat content of a system specifically at constant pressure.
28	B	A calorimeter is an insulated device to measure heat changes.
29	C	Heat of reaction is the enthalpy change for the written equation.
30	D	Molar heat of vaporization is heat absorbed as 1 mole vaporizes.
31	A	Standard heat of formation accompanies forming one mole of compound.
32	A	Heat (q) transfers due to temperature differences.
33	B	An exothermic process releases heat to surroundings.
34	B	Heat of combustion represents burning one mole.
Mat 36	D	An endothermic process absorbs heat from surroundings.

Question	Answer	Justification / Calculation
Mat 37	E	Heat capacity increases an object's temperature by 1°C.
Mat 38	F	The law of conservation of energy states energy is neither created nor destroyed.
Mat 39	B	The surroundings encompass everything outside the system.
Mat 40	A	The system is the part of the universe focused upon.
Mat 41	G	Calorimetry precisely measures heat flow.
Mat 42	H	A thermochemical equation includes enthalpy change.
Mat 43	I	Molar heat of condensation occurs during condensation.
Mat 44	J	Molar heat of fusion occurs during melting.
Mat 45	K	Molar heat of solidification occurs during freezing.
Mat 46	L	Molar heat of solution is caused by dissolution.
Mat 47	M	Hess's law allows summing equations and reaction heats.

Chapter 18: Reaction Rates and Equilibrium

Question	Answer	Justification / Calculation
1	D	Collision theory requires particles to hit with a specific minimum energy to react.
2	F	Reactant particles must possess sufficient energy to "climb over" the barrier.
3	E	Catalysts provide an alternative mechanism that inherently requires lower activation energy.
4	G	Enzymes are biological catalysts that speed up physiological processes.
5	G	Initial spark provides necessary collision energy; exothermic heat subsequently maintains it.
6	H	In reversible reactions, products continually form reactants while reactants continually form products.
7	F	Dynamic equilibrium occurs precisely when forward and reverse rates are equal.
8	E	Escaping CO ₂ forces equilibrium to shift, releasing bubbles to replace the lost gas.
9	E	$K_{eq} = \frac{[PCl_3][Cl_2]}{[PCl_5]}$ reflects products over reactants.

Question	Answer	Justification / Calculation
10	H	A large K_{eq} ($1 \times 10^2 > 1$) indicates that product formation is highly favorable.
11	G	Solids are omitted; $K_{sp} = [Ni^{2+}][S^{2-}]$.
12	E	K_{sp} mathematically is the product of ion concentrations raised to coefficient powers.
13	E	The lowest K_{sp} value denotes the least soluble salt.
14	D	According to Le Chatelier, adding a common ion shifts equilibrium left, lowering solubility.
15	F	Spontaneous reactions are characterized by a negative change in free energy.
16	G	Entropy is defined as the measure of disorder in a system.
17	H	An increase in order represents a decrease in entropy, making it unfavorable.
18	F	Solids possess a highly ordered crystalline structure, yielding lower entropy than liquids.
19	E	Two moles of gas forming one mole of solid results in a drastic decrease in entropy.
20	g	Endothermic processes can be spontaneous if accompanied by a highly favorable entropy increase.
21	B	Condensation is highly exothermic; this favorable enthalpy offsets unfavorable entropy.
22	D	Enthalpy change and entropy change together dictate the free energy (spontaneity) of a reaction.
23	A	It acts as a proportionality constant relating concentration to rate.
24	C	50 mins = 1 half-life (50% complete). 75% complete requires 2 half-lives ($50 \times 2 = 100$ mins).
25	D	The energy profile peak represents the activation energy barrier.

Chapter 19: Acids, Bases, and Salts

Question	Answer	Justification / Calculation
1	B	Arrhenius defined acids as substances ionizing to yield hydrogen ions in water.
2	A	Calcium hydroxide completely dissociates into Ca^{2+} and $2OH^-$ ions.

Question	Answer	Justification / Calculation
3	D	H ₂ O acts as the base by accepting a proton to form H ₃ O ⁺ .
4	C	Lewis theory covers compounds acting as bases/acids via electron pair donation/acceptance without H ⁺ exchange.
5	C	In neutral pure water, [H ⁺] and [OH ⁻] both equal 1.0×10^{-7} M.
6	C	pH = $-\log(1.0 \times 10^{-3}) = 3.00$; solutions below 7 are acidic.
7	A	pH + pOH = 14. pOH = 5. [OH ⁻] = $10^{-5} = 1.0 \times 10^{-5}$ M.
8	B	Strong acids dissociate completely, resulting in a large product-to-reactant ratio (large K _a).
9	C	$K_a = \frac{[\text{Products}]}{[\text{Reactants}]} = \frac{[\text{H}^+][\text{CN}^-]}{[\text{HCN}]}$.
10	A	Neutralization between an acid and a base yields a salt and water.
11	C	Visual indicators change color to signal the endpoint of the titration.
12	A	MaVa = MbVb. $0.50 \times 20.0 = M_b \times 40.0$. M _b = 0.25 M.
13	B	Weak-acid carbonate anions undergo hydrolysis, accepting water protons and increasing hydroxide/pH.
14	B	Once the buffer capacity is exceeded, pH begins to change significantly.
15	C	Amphoteric substances can act as both proton donors and acceptors.
16	B	A water molecule minus H ⁺ leaves OH ⁻ (hydroxide ion).
17	C	"Tri-" denotes three ionizable protons.
18	C	K _w is the ion-product constant for water (1×10^{-14}).
19	A	Buffer capacity defines the limit of addition before significant pH shifts occur.
Mat 1	H	A neutralization reaction produces salt and water.
Mat 2	I	Self-ionization describes two water molecules producing ions.
Mat 3	J	A conjugate base remains after an acid donates H ⁺ .
Mat 4	C	A Lewis acid accepts an electron pair.
Mat 5	D	Equivalence point is when H ⁺ moles equal OH ⁻ moles.
Mat 6	B	Titration uses known concentrations to find unknown ones.

Question	Answer	Justification / Calculation
Mat 7	G	Strong bases completely dissociate in aqueous solution.
Mat 8	A	Salt hydrolysis involves dissociated ions exchanging H ⁺ with water.
Mat 9	F	A conjugate acid forms when a base gains H ⁺ .
Mat 10	E	pH is the negative logarithm of the hydrogen-ion concentration.

Chapter 20: Oxidation-Reduction Reactions

Question	Answer	Justification / Calculation
1	C	Oxidation and reduction must always occur simultaneously.
2	A	A reducing agent donates electrons, thus undergoing oxidation.
3	B	Balanced: $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$. Mg goes from 0 to +2 (oxidized).
4	B	Gaining electrons (2e^- on reactant side) strictly defines reduction.
5	C	Sodium (Na) has low electronegativity and highly tends to lose electrons (acts as reducing agent).
6	A	Zn goes from 0 to +2 (oxidized/reducing agent); H goes from +1 to 0 (reduced/oxidizing agent).
7	C	Oxidation number assumes bonds are assigned to the more electronegative element.
8	C	The oxidation number of oxygen is almost always -2, not +2.
9	D	$\text{K} = +1$, $\text{O} = -2(4) = -8$. To maintain neutral charge, Mn must be +7.
10	A	For the sulfate ion (SO_4^{2-}), O is -2. $-2(4) = -8$. Overall charge is -2, so S = +6.
11	B	Sn^{2+} increases to Sn^{4+} (oxidized); Fe^{3+} decreases to Fe^{2+} (reduced).
12	C	Proper stoichiometric balancing yields $2\text{Na(s)} + \text{Cl}_2\text{(g)} \rightarrow 2\text{NaCl(s)}$.
13	B	Carbon and oxygen elements (both 0) combine to form a compound, necessitating oxidation states changes.
14	A	The half-reaction method in basic solution balances O with water and H with OH ⁻ , producing $3\text{CN}^- + 2\text{MnO}_4^- + \text{H}_2\text{O} \rightarrow 3\text{CNO}^- + 2\text{MnO}_2 + 2\text{OH}^-$.
15	B	Reduction is the gain of electrons or the loss of oxygen.

Question	Answer	Justification / Calculation
16	C	The oxidizing agent itself accepts electrons.
17	D	Half-reactions show isolated oxidation or reduction steps.
18	B	The oxidation-number-change method directly compares number increases and decreases.
Mat 50	C	Oxidation is the loss of electrons or gain of oxygen.
Mat 51	F	Oxidation-reduction reactions transfer electrons between reactants.
Mat 52	I	An oxidizing agent accepts electrons and is reduced.
Mat 53	A	Redox is short for oxidation-reduction.
Mat 54	E	A reducing agent donates electrons and is oxidized.
Mat 55	H	Reduction is the gain of electrons or loss of oxygen.
Mat 56	G	Oxidation number indicates oxidation/reduction degree.
Mat 57	B	A half-reaction shows isolated oxidation or reduction.
Mat 58	J	The half-reaction method balances components separately.
Mat 59	D	The oxidation-number-change method compares numerical increases/decreases.

Chapter 21: Electrochemistry

Question	Answer	Justification / Calculation
1	B	Silver is less reactive than zinc, resulting in no spontaneous redox reaction.
2	B	Magnesium sits highest among these options on the activity series.
3	A	Reduction (gain of electrons) inherently happens at the cathode.
4	C	The bridge maintains electrical neutrality via ion flow without fluid mixing.
5	D	A standard dry cell utilizes a zinc outer anode and a central carbon/graphite rod cathode.
6	A	Discharging consumes dense sulfuric acid and yields water, lowering solution density.
7	B	The active solid materials are Pb and PbSO ₄ at the anode, and PbO ₂ and PbSO ₄ at

Question	Answer	Justification / Calculation
		the cathode.
8	A	Hydrogen fuel cell operation remains less economical than conventional energy generation.
9	D	Fuel cells operate continuously as long as fuel is supplied, creating only water as a byproduct.
10	B	The standard hydrogen electrode was arbitrarily assigned 0.00 V to act as a universal reference.
11	C	Electric potential measures the cell's underlying ability to produce current.
12	A	Both tables sequence elements according to their relative tendencies to undergo redox processes.
13	C	An unknown half-cell potential is measured by connecting it to a standard hydrogen electrode.
14	B	A negative cell potential confirms the reaction is nonspontaneous and needs electrical input.
15	C	Cell Potential = Cathode - Anode. $+0.80\text{ V} - (-0.14\text{ V}) = +0.94\text{ V}$.
16	B	AC constantly changes direction, meaning metal deposited would be immediately stripped off.
17	A	The target object must be the cathode to attract positive silver ions from the electrolyte.
18	A	Electrolytic decomposition splits water into hydrogen and oxygen gases ($2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$).
19	C	The balanced equation produces twice as many moles (and volume) of hydrogen as oxygen.
20	B	The electrolysis of brine yields H_2 and Cl_2 in a 1:1 molar ratio.
21	A	Voltaic cells generate electricity chemically, whereas electrolytic cells consume electricity to force reactions.
22	A	Oxidation/reduction cannot occur in isolation; current flow demands both an electron source and destination.

Chapter 25: Nuclear Chemistry

Question	Answer	Justification / Calculation
1	B	An alpha particle is a helium nucleus; losing it decreases mass by 4 and atomic number by 2.
2	A	A radioisotope is simply an isotope that is radioactive.
3	B	Depleted rods cannot sustain the necessary reaction rate for constant electrical output.
4	B	Fusion demands extremely high temperatures that are difficult to achieve and contain.
5	B	Fission is initiated by neutron bombardment, splitting nuclei and releasing energy.
6	B	Natural radioactivity occurs in nature; artificial is generated in reactors/accelerators.
7	A	Ionizing radiation has enough energy to literally knock electrons off atoms.
8	B	$^{226}_{88}\text{Ra} \rightarrow ^{222}_{86}\text{Rn} + 4_2\text{He}$. (Corresponding to option B's balanced numbers).
9	A	Beta emission produces an electron: $^{14}_6\text{C} \rightarrow ^{14}_7\text{N} + 0_{-1}\text{e}$.
10	C	Gamma rays are pure high-energy photons lacking mass and charge.
Mat 11	B	Fission splits a nucleus into smaller fragments.
Mat 12	C	Fusion combines nuclei to produce a greater mass.
Mat 13	A	Transmutation converts an atom into a different element.
Mat 14	D	A beta particle is an electron from a broken neutron.
Mat 15	E	A gamma ray is a high-energy photon.